FRI Natural Language Processing Course 2026
Project report

University *of Ljubljana*
Faculty *of Computer and Information Science*

Report 2
Domain-Specific Fitness NLP Assistant Using Retrieval-Augmented Generation

Domen Pahole (Team leader), Luka Rizman, Nina Trivić

Abstract

This project develops a domain-specific NLP assistant for fitness applications. The system leverages structured exercise data to enable semantic retrieval and contextual response generation. This report presents the first implementation results and a comparison between a baseline LLM and a RAG-based approach.

Keywords

Natural Language Processing, Fitness NLP, Retrieval-Augmented Generation, Semantic Retrieval, Domain Adaptation

Advisors: Aleš Žagar

Introduction

In this report, we will introduce first results of testing our own RAG model in comparison with Llama 3.1 8B model. The topic is Fitness Assistant.

Datasets for Current Testing

For the current phase of testing, we use a single dataset, `MegaGymDataset.csv`. The dataset contains 2,918 exercise entries with structured metadata describing each exercise.

Each entry includes the following attributes:

- **Title** – name of the exercise
- **Desc** – textual description of how the exercise is performed
- **Type** – category of exercise (e.g. Strength, Cardio)
- **BodyPart** – primary muscle group targeted
- **Equipment** – required equipment
- **Level** – difficulty level (e.g. Beginner, Intermediate, Expert)
- **Rating** – numerical user rating
- **RatingDesc** – textual interpretation of the rating

An example entry from the dataset:

- **Title**: Dumbbell chest fly
- **Desc**: Description of the exercise execution
- **Type**: Strength
- **BodyPart**: Chest
- **Equipment**: Dumbbell
- **Level**: Intermediate

Results

In this section we present the first evaluation of our prototype RAG system. The goal is not a full-scale scientific evaluation, but rather to verify whether the core components already provide meaningful improvements over a baseline LLM setup.

We evaluate two aspects separately:

- Retrieval quality (vector search in Qdrant)
- End-to-end answer quality (Baseline LLM vs RAG)

Retrieval Evaluation

We first evaluate only the retrieval component, without using the LLM. The goal is to check whether the system can find relevant exercises for a given query.

Each test query is annotated with expected properties such as:

- target body part
- equipment
- optional difficulty level
- expected keywords

The retrieval pipeline is:

1. question → embedding (MiniLM)
2. embedding → Qdrant
3. top-5 most similar exercises returned
4. evaluation against expected criteria

Metrics

We use standard information retrieval metrics:

- Hit@k (whether a relevant result appears in top k)
- Precision@5
- Mean Reciprocal Rank (MRR)

Results

Metric	Value
Number of queries	12
Hit@1	0.500
Hit@3	0.917
Hit@5	0.917
Precision@5	0.583
MRR	0.694

Interpretation

The results show that the retrieval component already performs reasonably well.

Hit@5 = 0.917 means that in 11 out of 12 queries, at least one relevant exercise appears in the top 5 results. This is sufficient for a RAG system, where the LLM can use multiple candidates as context.

However, Hit@1 = 0.500 indicates that the top result is not always optimal. This is expected at this stage, as we use a general-purpose embedding model and datasets with inconsistent metadata.

A notable failure case is a query for beginner bodyweight leg exercises, where retrieval fails to return relevant results. This highlights limitations of:

- missing or inconsistent dataset fields
- lack of filtering by metadata (e.g. equipment)

Overall, retrieval is already usable, but clearly the main bottleneck for further improvement.

Baseline LLM vs RAG

We compare two approaches using the same model (Llama 3.1 8B via Ollama):

- Baseline: direct question → LLM
- RAG: question → retrieval → LLM with context

The difference lies only in the prompt, not in the model.

Key Observations

Aspect	Baseline	RAG
Knowledge source	General	Dataset + LLM
Exercise names	Generic	More specific
Grounding	None	Dataset-grounded
Hallucination risk	Higher	Lower (if retrieval good)
Weakness	Not dataset-aligned	Depends on retrieval

Example Insights

For queries such as:

"What chest exercises can I do with dumbbells?"

the baseline produces reasonable general answers, but may include exercises not present in our dataset.

The RAG system, on the other hand, generates answers based on retrieved entries, leading to:

- more consistent exercise naming
- better alignment with the dataset
- structured explanations based on retrieved metadata

However, in cases where retrieval fails (e.g. broad queries like beginner bodyweight legs), the RAG answer becomes worse than the baseline. This confirms that RAG performance is highly dependent on retrieval quality.

Discussion

The comparison highlights an important insight:

- Baseline LLM is strong in general reasoning and common knowledge
- RAG is strong in grounding answers to a specific dataset
- The overall system quality is dominated by retrieval performance

This suggests that the next development focus should not be a larger model, but improvements in retrieval:

- metadata filtering (BodyPart, Equipment, Level)
- dataset cleaning and deduplication
- improved evaluation dataset

Conclusion of Current Phase

At this stage, we have successfully implemented a working RAG prototype:

- vector-based retrieval using Qdrant
- embedding model integration
- prompt-based grounding for LLM
- initial evaluation framework

The system already demonstrates the core advantage of RAG: grounding responses in a controlled dataset.

The main limitation is retrieval quality, which directly affects final answer quality. This will be the primary focus of further work.

References